Subiect 1, Calculatoare

(*A*) $f(x) = \ln \frac{1+x}{1-x} + 2\arctan x$.
$f^{(n)}(0) = ?$

(*B*) Prove that in a topological space if a set ✤ is open, then ✤ $\subseteq$ int(✤).

(*C*) Prove that in a topological space ✤$'$ $\subseteq \overline{✤}$ or viceversa.

Subiect 2, Calculatoare

(*A*) $✤(x) = \sin x \sim$ **Maclaurin**

(*B*) **Find the first decimal digit of** $\sin 1$.

(*C*) $\lim\limits_{n\to\infty} \sqrt[n]{\sqrt[n]{✤_n^2}} = ?$

Subiect 3, Calculatoare

(*A*) Let $\clubsuit_n > 0$ such that $\lim\limits_{n\to\infty} \left(\frac{\clubsuit_{n+1}}{\clubsuit_n}\right)^n < 1$. Prove that $\lim\limits_{n\to\infty} \clubsuit_n = 0$.
Give an example, not from within the course.

(*B*) Let (X, d) be a metric space and $\clubsuit\colon X \to X$ be a contraction. Prove that $\clubsuit$ has at most one fixed point.

(*C*) In the natural topology of the real axis find $\mathbb{N}'$.

Subiect 1, Automatica

(*A*) Prove, in three ways, that the series $\sum_{n=1}^{\infty} \frac{1}{n^{\circledast}}$, $\circledast > 1$, is convergent.

(*B*) Find limit of the sequence defined by

$$\circledast_0 \in \mathbb{N}^*, \quad \circledast_{n+1} = \begin{cases} \dfrac{\circledast_n}{2}, & \text{if } \circledast_n \text{ is even,} \\ \dfrac{1+\circledast_n}{2}, & \text{if } \circledast_n \text{ is odd,} \end{cases} \qquad n = 0, 1, \ldots \; .$$

Subiect 2, Automatica

(*A*) Find an alternating series for $\log 5$.

(*B*) Prove Taylor's Formula with Integral Form of the Remainder

$$\clubsuit(x) = \clubsuit(a) + (x-a)\clubsuit'(a) + \int \cdots$$

(*C*) The nature of the series $\displaystyle\sum_{n=2}^{\infty} \frac{\log n!}{n^{\clubsuit}}$, $\clubsuit > 2$.

Subiect 3, Automatica

(*A*) Let $\text{⌘}_i(x) = \sum_{k=1}^{n} a_{i,k}\, x^k,\ i = 0, \ldots, n,\ x \in \mathbb{R}$. **Calculate the determinant**

$$\begin{vmatrix} \text{⌘}_0(x) & \text{⌘}_1(x) & \cdots & \text{⌘}_n(x) \\ \text{⌘}_0'(x) & \text{⌘}_1'(x) & \cdots & \text{⌘}_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ \text{⌘}_0^{(n)}(x) & \text{⌘}_1^{(n)}(x) & \cdots & \text{⌘}_n^{(n)}(x) \end{vmatrix}, \qquad x \in \mathbb{R}. \ ✓$$

(*B*) Find the equation of the tangent plane at the point a **of the surface given by** $\text{⌘}(x) = 0$.

(*C*) Find the first decimal digit of $\displaystyle\int_0^1 e^{-x^4}\,\mathrm{d}x$.